

Project Update 6

ECON 345

Due: 11:59PM October 31, 2026

For this assignment, you will need to submit:

1. Your RMarkdown file (**download and use the template we provided, `project_hw6_student.Rmd`**)
2. A knitted PDF.

If you cannot knit to a PDF, type your answers into a Word document (remember to number them accordingly), and be sure to include screenshots of your outputs for each question. Submit this Word document together with your `.Rmd` file so we can later help troubleshoot your problem with knitting.

Part A - Iterating on Previous Work (30 points)

For this assignment, you need to reproduce your work from Project Update 5 with revisions based on feedback you received. If you did not receive any feedback for a particular item, you should still include that item.

1. **Research Question (10 points)**
 - What is your research question? (2)
 - How has your research question changed over the course of the project? (2)
 - What are the key relationships in your data that could inform the answer to your research question? (2)
 - Based on your data, what is a testable hypothesis for your research question? (4)
2. **Summary Statistics (4 points)**
 - Using the `vtable` or `kable` package and incorporating weights (if your project has weights), compute and present summary statistics of your dependent and independent variables. (2)
 - If you received feedback on your summary statistics or their interpretation, state the feedback and then incorporate it here. If there's anything you're struggling to incorporate, state that. If you had no feedback for this item, state that. (2)
3. **Conditional Summary Statistics (4 points)**
 - Using the `vtable` or `kable` package and incorporating weights (if your project has weights), compute and present conditional summary statistics of your dependent variable by levels/groups of your independent variable(s). (2)
 - If you received feedback on your conditional summary statistics or their interpretation, state the feedback and then incorporate it here. If there's anything you're struggling to incorporate, state that. If you had no feedback for this item, state that. (2)
4. **Data Visualization (12 points)**
 - Create three charts that are key to telling your story. At least one of these should show a key relationship between two separate variables (e.g., a conditional histogram or scatterplot). Refer to the data visualization checklist for tips on making your charts easy to read and informative. (5)
 - For each chart, describe the part of the story it tells. (5)
 - If you received feedback on your data visualizations or their interpretation, state the feedback and then incorporate it here. If there's anything you're struggling to incorporate, state that. If you had no feedback for this item, state that. (2)

Part B - Refining Your Interpretation (30 points)

5. What did you originally expect to find in the data about the relationship between your key variables? (5)
6. How have your findings differed from your original expectations? (5)
7. How does your analysis thus far help inform your research question? Tell us a story. (10)
8. What limitations exist to your analysis and visualizations so far? How could you overcome these limitations? (10)

Part C - Analysis With a Simple Regression (20 points)

9. Define your dependent variable and one explanatory variable for a simple regression. Explain your reasoning — why do you expect a relationship between them, and what sign do you expect? You may want to use the same variables as in Question 4. (5)
10. State the null and alternative hypotheses for the slope on your explanatory variable. (5)
11. Fit the regression with `lm()`, and display the results with `broom::tidy()` and `broom::glance()`. **Do not use `summary()` or pull coefficients with `$`** — this course uses the tidy workflow for regression output. (5)
12. Interpret the results: report the coefficient and what it means in the units of your data, note whether it's statistically significant, and discuss whether it's economically significant given your summary statistics. Visualize the relationship with a scatterplot and the fitted line from `broom::augment()`. (5)

Part D - Follow-up With Your Adviser (10 points)

13. Email your adviser with your knitted output (the PDF or Word document that results from your RMarkdown file). Do not send your R script or your `.Rmd`, but do submit those on Canvas.
 - Email your knitted output to your adviser, with `econ345.spelman@gmail.com` cc'd. (5)
 - In your email, give your adviser a brief summary of what you've done this week and what you've learned about your research question. Include any questions or ideas about where to take your analysis next. (5)

Part E - Share With Your Classmates (10 points)

14. Pick one chart you particularly like from Part A and share it with your classmates on the Canvas discussion board, with a brief description of what it means.
 - Posting your chart (5)
 - A brief description of your chart (5)

You're strongly encouraged to comment on your classmates' posts as well!